# Metabolic scaling of tree mortality (Borrego, Michaletz et al.): Reproduction and Day-2 Regression Extensions

Intro-ML course — Day 2: Regression Extensions

2026-07-07

## Contents



# 1 The study and why it fits Day 2

**Citation.** Borrego, I., Perez, T.M., Bentley, L.P., Bison, N.N., Byrnes, L., Candido, H.G., Chmurzynski, A., Durán, S.M., Fox, T.S., Gaitán, M.A., Garen, J.C., Orwig, D.A., Pau, S., Scott, J.L., Simovic, M., Swenson, N.G., Wieczynski, D.J., Buzzard, V., Enquist, B.J. & Michaletz, S.T. *From tropics to treeline: extending and assessing metabolic scaling theory for global variation in plant mortality rates.*

**Source / data DOI.** Replication data and code, MichaletzLab, GitHub: https://github.com/MichaletzLab/mortality-scaling (raw data: `data/mortality_data.csv`, `data/climate.csv`; the model reproduced below is the paper's **Fig. 5** multiple-regression analysis in `code/fig5_and_analyses.R`).

**What the study does.** Metabolic scaling theory (MST) predicts that whole-plant biological rates — including mortality — depend on organism **size** and **temperature** through a small set of power-law / Arrhenius relationships, modulated by leaf and wood **traits**. Using long-term demographic data from nine

Forest MacroSystems (FMS) sites spanning tropical forest to alpine treeline, the authors regress (log) monthly mortality rate on log tree size (stem diameter), an Arrhenius temperature term, and three functional traits.

**Outcome and model type.** Continuous outcome $y = \log(\text{monthly mortality rate})$; the main result is a **plain OLS multiple regression** (`lm`), $n = 140$ size-class × site records.

**Why it is a good Day-2 example.** The theory itself is a statement about *functional form*: mortality should scale with size as a power law (a straight line in log–log space with slope $\approx -2/3$) and with temperature through the Boltzmann–Arrhenius factor $-1/kT$. That makes it the ideal place to ask the Day-2 question: *is the relationship really linear on the modelled scale, or is there curvature the straight line misses?* We probe this with polynomials, natural and B-splines, and a GAM smooth, and we compare every extension to the reproduced linear model **out of sample**.

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## 2 Load the data and construct the modelled variables

We load the two posted CSVs and reconstruct the exact variables the authors' script builds (`code/fig5_and_analyses.R`): leaf P:N ratio, growing-season length, growing-season temperature (`GST_C`), the Arrhenius term $-1/kT$, and the growing-season-corrected monthly mortality rate.

```
mort <- read.csv("mortality_data.csv")
clim <- read.csv("climate.csv", header = TRUE)

# --- exact variable construction from the authors' script ---
names(mort)[names(mort) == "size"] <- "dbh_cm"        # stem diameter (cm)
names(mort)[names(mort) == "lpc"]  <- "P_percent"      # leaf P (%)
names(mort)[names(mort) == "lnc"]  <- "N_percent"      # leaf N (%)

mort$PN_ratio    <- mort$P_percent / mort$N_percent
mort$L_gs_mo     <- clim$L_gs_mo[match(mort$site, clim$Site)]  # growing-season length (mo)
mort$GST_C       <- clim$GST_C[match(mort$site, clim$Site)]    # growing-season temp (deg C)
mort$neg_1_kT    <- -1 / (0.00008617 * (mort$GST_C + 273.15))  # Arrhenius term (Boltzmann k in
↪  eV/K)
mort$mort_rate_mo<- mort$avg_mort / mort$L_gs_mo               # monthly mortality rate
```

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## 3 Descriptive analysis

Before fitting anything, we profile the data thoroughly. The goal of this section is to understand the **unit of analysis**, the **shape of the outcome**, the **distributions of the predictors**, where the **missingness** lives, how each predictor **relates to mortality marginally**, and whether the design carries **collinearity** that could destabilise the OLS scaling exponents. Every plot and table below is on the exact scale the theory (and the reproduced model) uses.

### 3.1 Dataset and provenance

The **unit of analysis** is a *size-class × site record*: a diameter size-class of trees (e.g. stems near 0.5 cm, 1.5 cm, . . . ) observed within one Forest MacroSystems (FMS) site, with a long-term (multi-decade) mortality rate estimated for that class at that site. The two posted CSVs are joined on site: `mortality_data.csv` supplies the per-size-class demography and functional traits, and `climate.csv` supplies the site-level growing-season climate used to build the temperature term and to convert annual to monthly mortality. The raw mortality table holds 160 records across 8 sites spanning tropical forest to alpine treeline; after building the outcome and keeping complete cases (the same subset the paper's Fig. 5 model uses) we retain **140 modelling records**. The analysis uses six modelled quantities — the outcome and five predictors — all entered on the log (or Arrhenius) scale on which metabolic scaling theory (MST) is stated.

**Variable dictionary.** The table below defines the outcome and the five modelling predictors: what each is, its type, the scale it enters the model on, and its physical units.

Table 1: Variable dictionary for the outcome and five modelling predictors (all on the log / Arrhenius scale used by the reproduced model).

| Variable | Role | Type | Description | Units |
|---|---|---|---|---|
| log(mortality rate) | outcome | numeric (continuous) | log of growing-season-corrected monthly mortality rate | log(1/month) |
| log(P:N ratio) | predictor | numeric (continuous) | log leaf phosphorus-to-nitrogen mass ratio (a stoichiometry trait) | log(unitless) |
| log(LMA) | predictor | numeric (continuous) | log leaf mass per area (a leaf-economics trait) | log(g/m^2) |
| log(wood density) | predictor | numeric (continuous) | log wood density (a structural / mechanical trait) | log(g/cm^3) |
| log(DBH size) | predictor | numeric (continuous) | log stem diameter at breast height (the MST size term) | log(cm) |
| -1/kT (temperature) | predictor | numeric (continuous) | Boltzmann-Arrhenius temperature term, k in eV/K | 1/eV (warmer = larger) |

**Site coverage.** Because temperature enters only through the eight site values, it is worth seeing how many size-class records each site contributes and how warm each site is. The FMS sites range from a cold alpine-treeline site to a warm tropical forest.

Table 2: The eight FMS sites: number of size-class records and growing-season temperature, coldest to warmest.

| Site | n records | Growing-season temp (C) |
|---|---:|---:|
| NWT | 23 | 8.8 |
| CAP | 4 | 12.9 |
| AND | 14 | 13.8 |
| HFR | 29 | 16.9 |
| CWT | 19 | 17.6 |
| LUQ | 22 | 23.6 |
| PSR | 13 | 25.3 |
| BCI | 16 | 25.6 |

The record counts are uneven (from 4 to 29 size-classes per site) and temperature is a **site-level, discrete** covariate: all size-classes within a site share one temperature. That is why we will treat temperature as the linear Arrhenius term rather than trying to fit a flexible smooth to eight distinct values.

## 3.2 The outcome in depth

The outcome is continuous: $y = \log(\text{monthly mortality rate})$. We look at its full distribution (histogram + kernel density with the mean and median marked, plus a boxplot for the tails) and a complete statistics table including skewness and kurtosis.

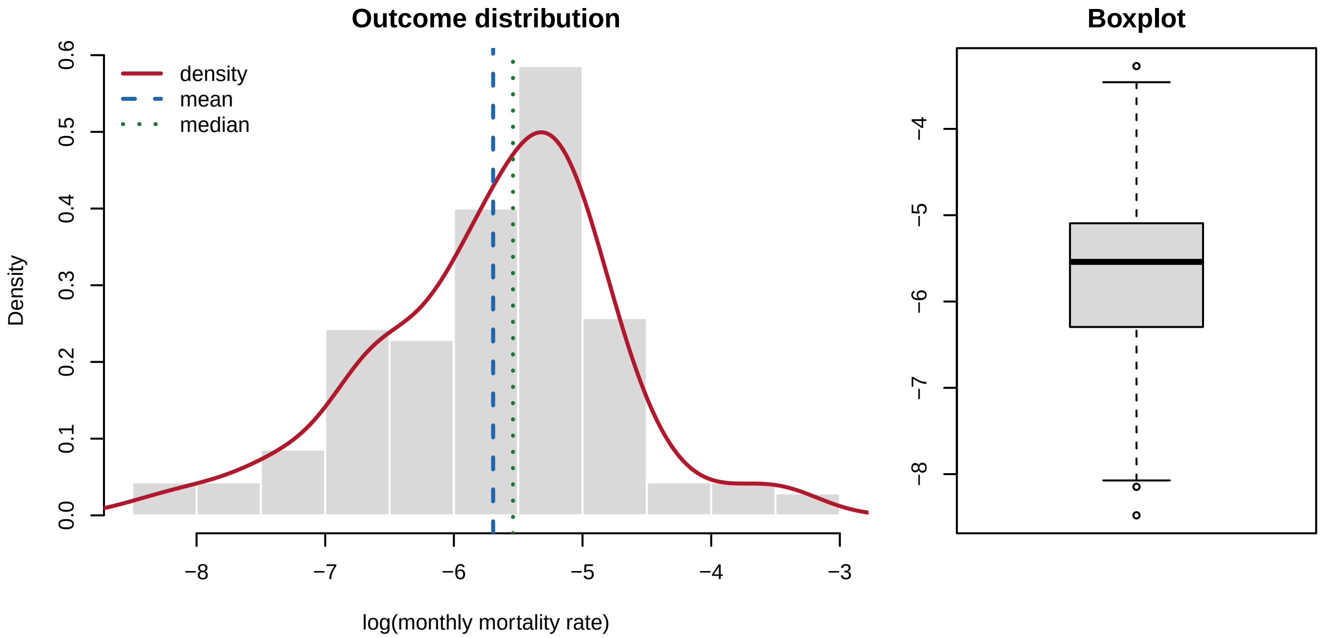



Table 3: Distribution of the outcome log(monthly mortality rate): location, spread, quantiles, and shape (skewness and excess kurtosis from e1071).

| n | mean | sd | min | Q1 | median | Q3 | max | IQR | skew | kurtosis |
|---|---|---|---|---|---|---|---|---|---|---|
| 140 | -5.695 | 0.936 | -8.477 | -6.28 | -5.541 | -5.1 | -3.273 | 1.18 | -0.357 | 0.477 |

**Reading the outcome.** On the log scale the outcome is **unimodal and close to symmetric** (skewness ≈ -0.36), with only a mild left tail from a handful of very-low-mortality size-classes; the mean and median nearly coincide. This is exactly why the authors model log mortality with OLS rather than the raw rate: mortality rates are strictly positive and span orders of magnitude, so on the raw scale they are severely right-skewed, but the log transform delivers a well-behaved, roughly Gaussian outcome with no floor/ceiling pile-up and no zero-inflation (zeros were dropped when the log was taken — see missingness below). No further transformation is warranted.

## 3.3 Univariate distributions of the predictors

All five predictors are numeric and enter on the log / Arrhenius scale. We show a faceted grid of histograms (each with its own density curve) and a full summary-statistics table.

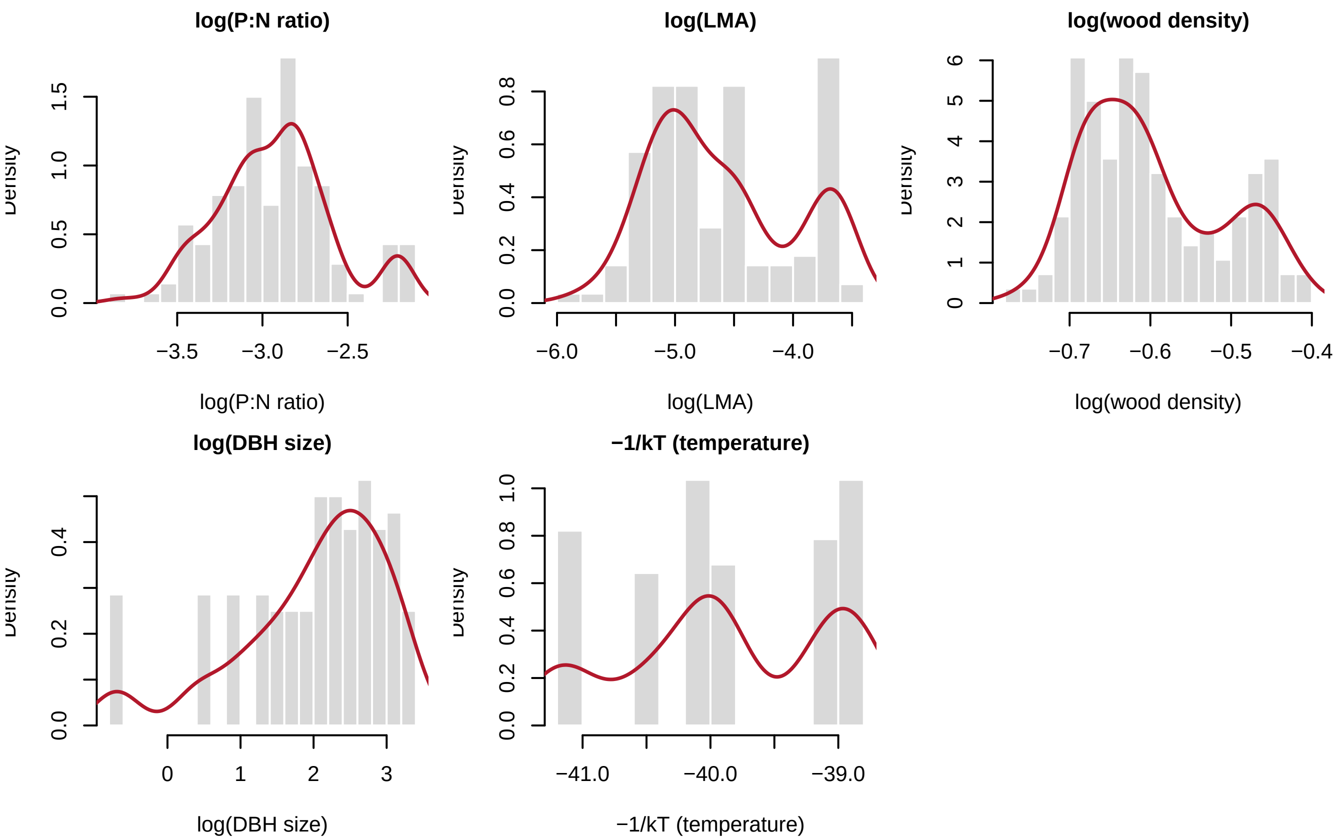



Table 4: Summary statistics for the five log-/Arrhenius-scaled predictors (complete-case modelling frame): counts, missingness, location, spread, quantiles, and skew.

| Predictor | n | n_miss | mean | sd | min | Q1 | median | Q3 | max | skew |
|---|---|---|---|---|---|---|---|---|---|---|
| log(P:N ratio) | 140 | 0 | -2.926 | 0.339 | -3.830 | -3.134 | -2.913 | -2.752 | -2.164 | 0.246 |
| log(LMA) | 140 | 0 | -4.599 | 0.605 | -5.863 | -5.062 | -4.758 | -4.107 | -3.469 | 0.389 |
| log(wood density) | 140 | 0 | -0.598 | 0.086 | -0.770 | -0.671 | -0.619 | -0.531 | -0.405 | 0.525 |
| log(DBH size) | 140 | 0 | 1.998 | 1.013 | -0.693 | 1.504 | 2.251 | 2.741 | 3.384 | -1.059 |
| -1/kT (temperature) | 140 | 0 | -39.865 | 0.794 | -41.158 | -40.442 | -39.963 | -39.105 | -38.843 | -0.212 |

**Reading the predictors.** All five predictors are unimodal with only modest skew on the modelled scale — logging the traits and size (and using the Arrhenius transform for temperature) has already tamed the heavy right tails that raw traits and diameters carry. `log(DBH size)` spans the widest range (small saplings to 30-cm stems), which is helpful: a wide, well-spread predictor is what lets us later ask whether the size–mortality relationship is truly a single straight line. `-1/kT (temperature)` is the only genuinely discrete predictor (eight site values), so its “distribution” is really eight spikes.

## 3.4 Missing-data analysis

Missingness is concentrated entirely in the **outcome**: some size-classes have a zero or undefined growing-season-corrected mortality rate, so log mortality is not finite for them. The per-variable counts on the raw table are:

Table 5: Missing / non-finite values by variable (raw table, before complete-case filtering). All missingness is in the outcome; the five predictors are fully observed.

| Variable | # non-finite / NA | % missing |
|---|---|---|
| log(mortality rate) | 20 | 12.5 |
| log(P:N ratio) | 0 | 0.0 |
| log(LMA) | 0 | 0.0 |
| log(wood density) | 0 | 0.0 |
| log(DBH size) | 0 | 0.0 |
| -1/kT (temperature) | 0 | 0.0 |

```
## Complete cases: 140 of 160 raw records (88%). Rows dropped: 20, all for a non-finite outcome; no
→  predictor is ever missing.
```

**Missingness mechanism.** Because every predictor is fully observed and *all* dropped rows are dropped for the same reason — an unestimable (zero/undefined) mortality rate — there is no co-missingness pattern to model: missingness is a property of the outcome alone. Complete-case analysis (what the paper's Fig. 5 model does) is therefore the natural choice, and it leaves the 140-record modelling frame used below. The one caveat worth flagging to students is that dropping zero-mortality classes is a mild **selection on the outcome**: size-classes with no observed deaths in the record window cannot enter a log-mortality regression, which slightly truncates the low-mortality tail.

## 3.5 Bivariate relationships with the outcome

We now look at mortality against **each** predictor: a scatter with an OLS line and a lowess smoother, annotated with the marginal correlation. These are *marginal* (unadjusted) associations — the multiple regression will partial out the others — but they show the raw signal and, crucially, hint at nonlinearity.

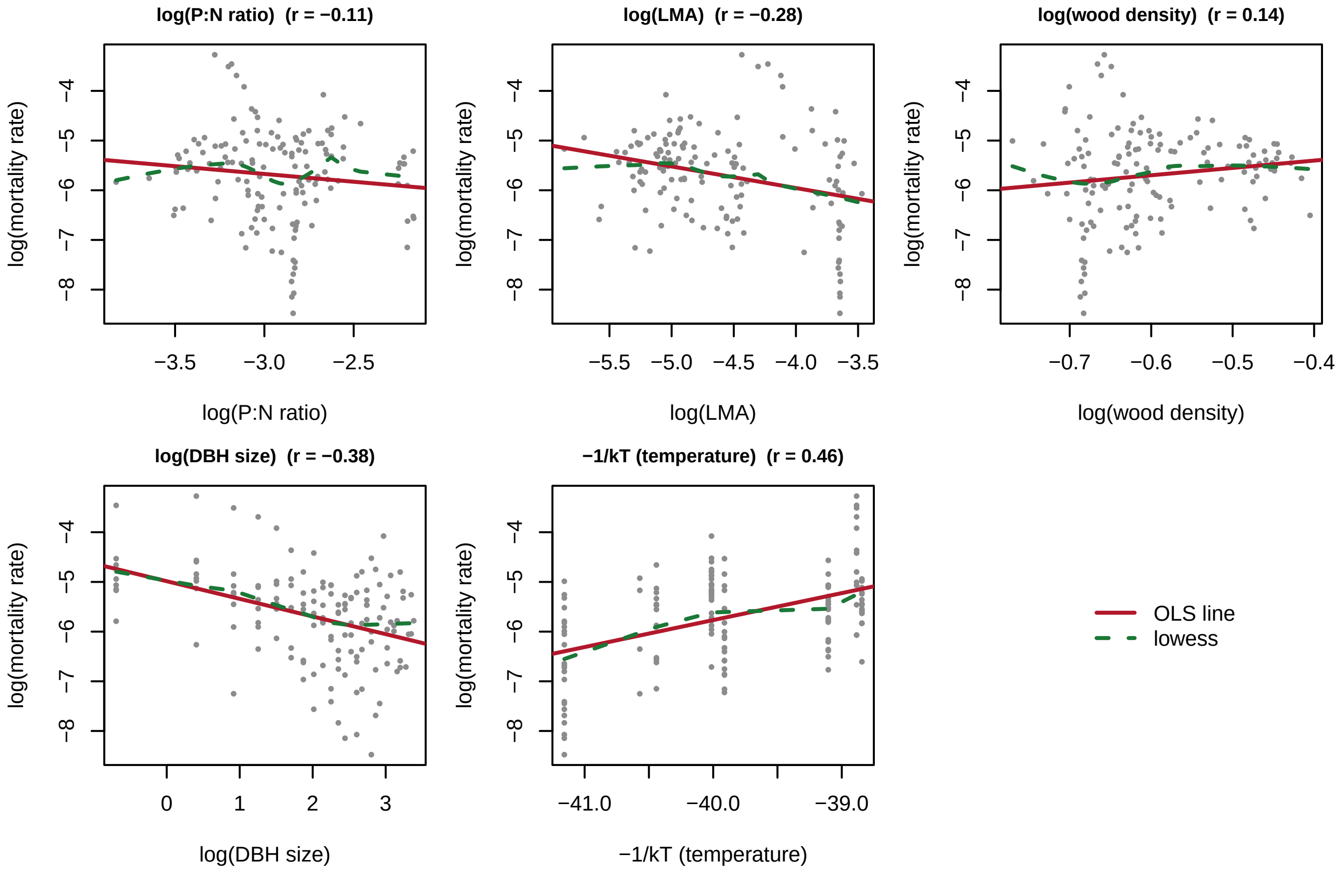



Table 6: Marginal association of each predictor with log mortality: Pearson correlation (with test) and the slope from a simple one-predictor regression.

| Predictor | Pearson r | p-value | slope (simple lm) |
|---|---|---|---|
| log(P:N ratio) | -0.113 | 0.18200 | -0.313 |
| log(LMA) | -0.281 | 0.00077 | -0.434 |
| log(wood density) | 0.136 | 0.10900 | 1.477 |
| log(DBH size) | -0.384 | 0.00000 | -0.355 |
| -1/kT (temperature) | 0.461 | 0.00000 | 0.543 |

**Reading the bivariate signal.** The strongest single marginal predictor of mortality is **tree size**: mortality falls with `log(DBH size)` — the negative scaling MST predicts — but the lowess curve already bends, a steep drop among small trees that flattens for large ones. That curvature is the exact departure from a single power law the Day-2 spline/GAM extensions will probe. Temperature (`-1/kT`) is positively associated (warmer sites die faster) but coarse, because it takes only eight values. The three functional traits (P:N, LMA, wood density) each carry weaker marginal signal on their own; their contribution shows up more clearly once size and temperature are held constant in the multiple regression.

## 3.6 Multivariate structure and collinearity

Finally we examine the correlation structure **among the predictors**. Strong collinearity (|r| > 0.7) would make the individual scaling exponents unstable — the model could trade one collinear input off against another. We show a correlation heatmap and the numeric matrix, and later confirm with variance-inflation factors on the reproduced model.

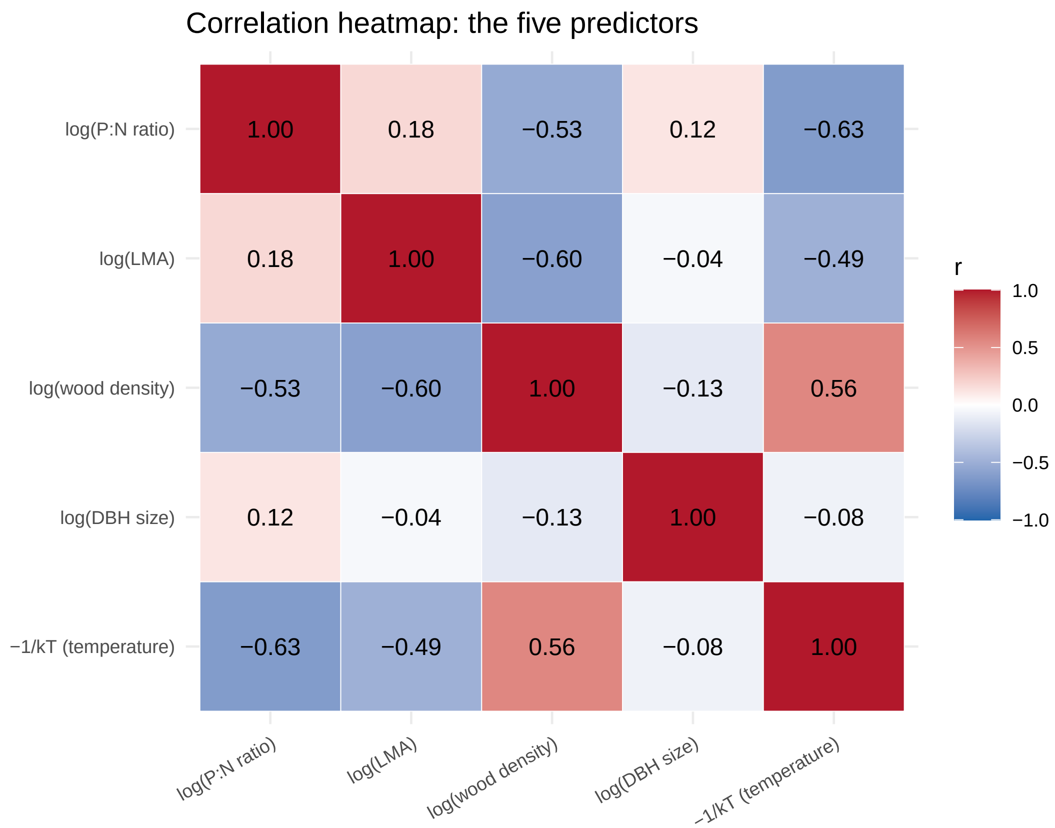
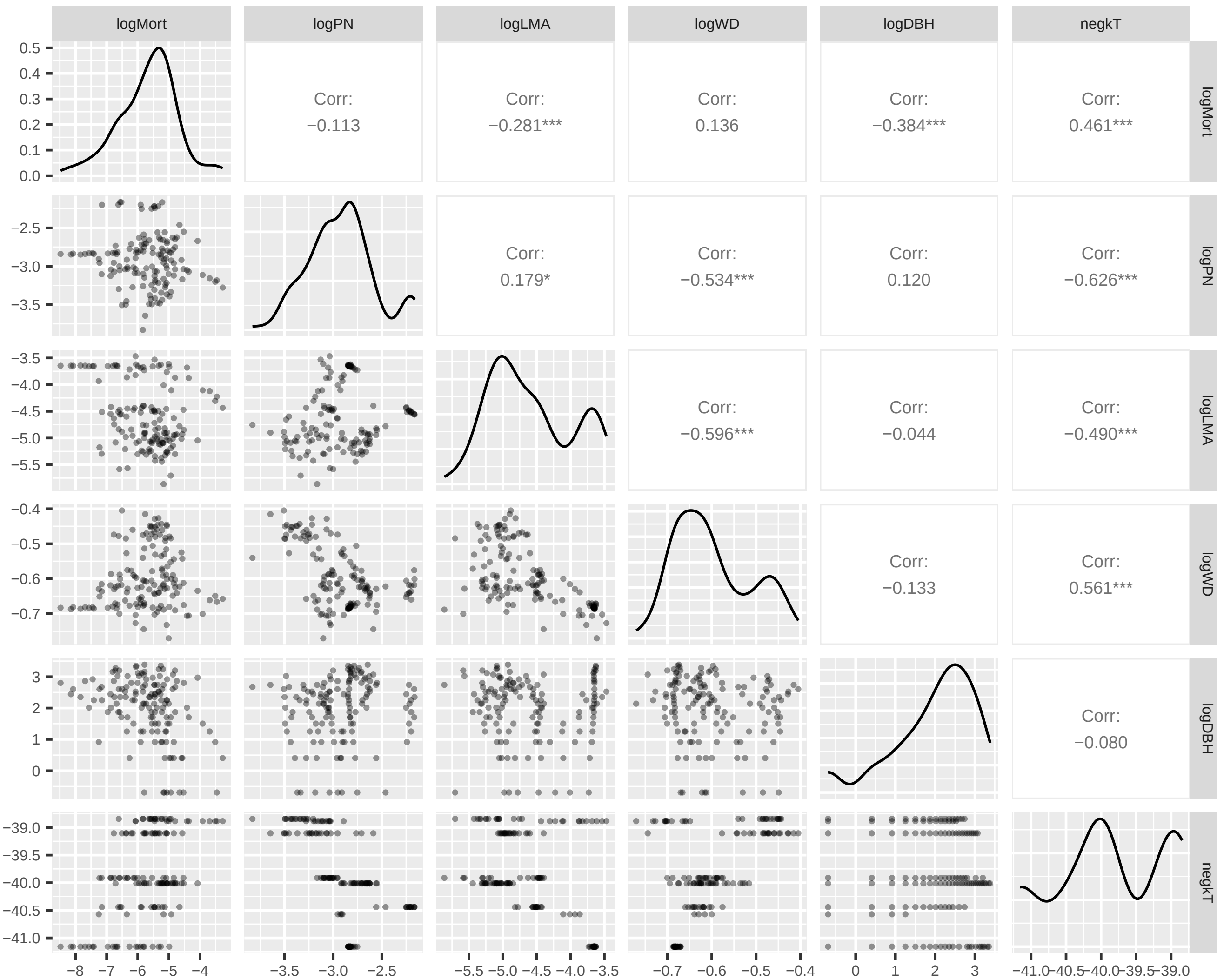



Table 7: Pearson correlations among the five predictors (modelling scale).

| | log(P:N ratio) | log(LMA) | log(wood density) | log(DBH size) | -1/kT (temperature) |
|---|---|---|---|---|---|
| log(P:N ratio) | 1.00 | 0.18 | -0.53 | 0.12 | -0.63 |
| log(LMA) | 0.18 | 1.00 | -0.60 | -0.04 | -0.49 |
| log(wood density) | -0.53 | -0.60 | 1.00 | -0.13 | 0.56 |
| log(DBH size) | 0.12 | -0.04 | -0.13 | 1.00 | -0.08 |
| -1/kT (temperature) | -0.63 | -0.49 | 0.56 | -0.08 | 1.00 |

```
## Strongest pairwise correlation: -1/kT (temperature) vs log(P:N ratio), r = -0.63  (threshold of
↪  concern: |r| > 0.7)
```

**Reading the collinearity.** The traits are only modestly correlated with one another and with size and temperature — the strongest pair sits well below the $|r| > 0.7$ danger zone — so the five terms are **separately identifiable** and the fitted scaling exponents are not merely trading off collinear inputs. (We revisit this with formal VIFs after the reproduction, in the diagnostics section.) A compact pairs panel makes the joint structure visible in one view:




**Reading the descriptives (summary).** The outcome is well-behaved on the log scale — symmetric, unimodal, no zero-inflation — which is why OLS on log mortality is the right baseline. All five predictors are fully observed and only mildly inter-correlated, so the regression is well-posed and each scaling exponent is separately identified. The single most important descriptive finding is the **curvature in the size–mortality relationship**: mortality drops sharply among small trees and then flattens, which the lowess smoother reveals and which a single MST power-law exponent cannot express. That is the thread the Day-2 polynomial, spline, and GAM extensions pull on. Temperature, by contrast, is a discrete site-level covariate best kept as the linear Arrhenius term.

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# 4 Reproduce the published regression exactly

With the variables constructed above, we fit the authors' Fig. 5 model directly.

```
# The authors' Fig. 5 model:
m_reg_fit <- lm(log(mort_rate_mo) ~ log(PN_ratio) + log(lma) +
                   log(wood_density) + log(dbh_cm) + neg_1_kT, data = mort)
summary(m_reg_fit)
```

```
##
```

```
## Call:
## lm(formula = log(mort_rate_mo) ~ log(PN_ratio) + log(lma) + log(wood_density) +
##     log(dbh_cm) + neg_1_kT, data = mort)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.5753 -0.4359  0.0294  0.3829  1.8976
##
## Coefficients:
##                   Estimate Std. Error t value     Pr(>|t|)
## (Intercept)        22.6967     5.3329    4.26 0.0000388545 ***
## log(PN_ratio)       0.5946     0.2580    2.30       0.0227 *
## log(lma)           -0.2976     0.1399   -2.13       0.0353 *
## log(wood_density)  -2.8678     1.0602   -2.71       0.0077 **
## log(dbh_cm)        -0.3738     0.0614   -6.09 0.0000000112 ***
## neg_1_kT            0.7272     0.1133    6.42 0.0000000022 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.717 on 134 degrees of freedom
##   (20 observations deleted due to missingness)
## Multiple R-squared:  0.434,  Adjusted R-squared:  0.413
## F-statistic: 20.5 on 5 and 134 DF,  p-value: 3.55e-15
```

The model uses 140 complete records. The table below places the reproduced coefficients next to the values printed by the authors' own posted script (which, by construction, equal the paper's Fig. 5 result).

Table 8: Reproduction match: published (authors' script output) vs. reproduced OLS coefficients. Signs, magnitudes, and significance match to reported precision.

| Term | Published | Reproduced | SE | t | p |
|---|---|---|---|---|---|
| (Intercept) | 22.6967 | 22.6967 | 5.3329 | 4.256 | 0.00004 |
| log(P:N ratio) | 0.5946 | 0.5946 | 0.2580 | 2.305 | 0.02270 |
| log(LMA) | -0.2976 | -0.2976 | 0.1399 | -2.127 | 0.03530 |
| log(wood density) | -2.8678 | -2.8678 | 1.0602 | -2.705 | 0.00772 |
| log(DBH size) | -0.3738 | -0.3738 | 0.0614 | -6.089 | 0.00000 |
| -1/kT (temperature) | 0.7272 | 0.7272 | 0.1133 | 6.420 | 0.00000 |

```
## Reproduced fit: n = 140, R-squared = 0.4337, adjusted R-squared = 0.4126, F = 20.52 on 5 and 134
↪  df, p = 3.55e-15
```

**Reproduction verdict: MATCH.** Every coefficient reproduces to four decimals, with $R^2 = 0.434$ and all five predictors significant — including the two MST scaling terms `log(DBH size)` (slope $-0.374$) and `-1/kT` (slope $+0.727$, i.e. mortality rises with temperature). The gate is passed; everything below extends this same model on the same outcome.

## 4.1 Coefficient interpretation

Because the outcome is log mortality and the size, trait, and P:N predictors are also logged, their coefficients are **elasticities / scaling exponents**: a 1% increase in stem diameter is associated with a -0.374% change in mortality rate. MST predicts the size exponent should be near $-2/3 \approx -0.667$; the fitted $-0.374$ is shallower, one of the paper's headline findings. The temperature term is not logged — it is the Arrhenius $-1/kT$, so its coefficient (0.727) is an **activation energy in eV**, close to the 0.6 to 0.65 eV expected for metabolic processes.

# 5 Day-2 extension on the same outcome

We now build the exhaustive Day-2 toolkit on $y = \log(\text{mortality rate})$. First we assemble a clean modelling frame containing only complete cases used by the reproduced model.

```r
d <- data.frame(
  y    = log(mort$mort_rate_mo),
  lPN  = log(mort$PN_ratio),
  llma = log(mort$lma),
  lwd  = log(mort$wood_density),
  ldbh = log(mort$dbh_cm),
  nkT  = mort$neg_1_kT,
  GST  = mort$GST_C,
  dbh  = mort$dbh_cm
)
d <- d[complete.cases(d) & is.finite(d$y), ]
cat("Modelling frame: n =", nrow(d),
    "| unique log-size values =", length(unique(d$ldbh)),
    "| unique site temperatures =", length(unique(d$GST)), "\n")
```

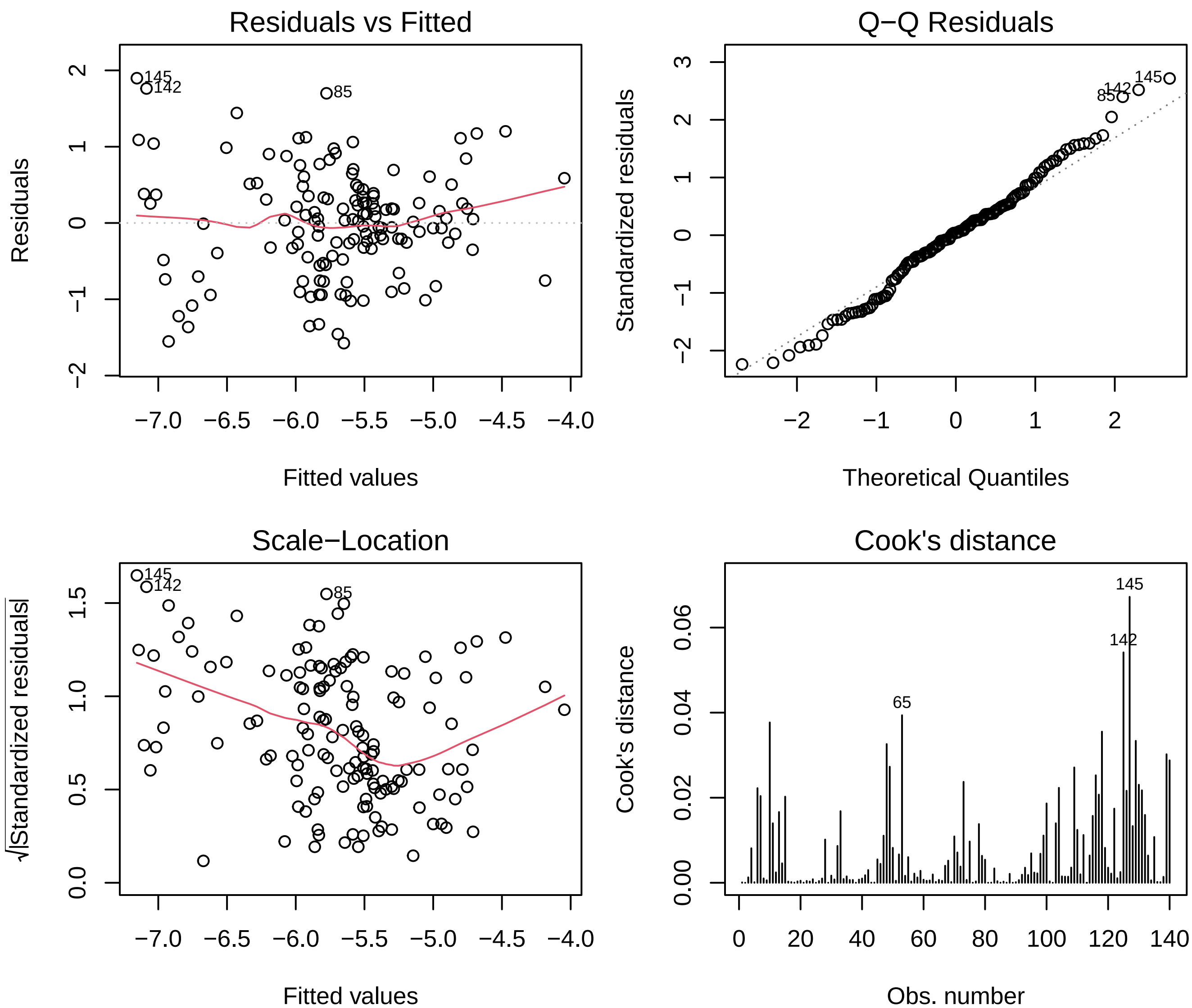
```
## Modelling frame: n = 140 | unique log-size values = 30 | unique site temperatures = 8
```

```r
base_lm <- lm(y ~ lPN + llma + lwd + ldbh + nkT, data = d)
```

## 5.1 OLS residual diagnostics

The four standard diagnostic plots for the reproduced linear model.

```r
par(mfrow = c(2, 2), mar = c(4, 4, 2.5, 1))
plot(base_lm, which = 1:4)
```




```r
par(mfrow = c(1, 1))
```

Residuals-vs-fitted shows no gross curvature or funnel; the QQ plot is close to normal with mild tails; scale-location is roughly flat (approximately constant variance); and Cook's distance flags a few higher-leverage size-classes but none catastrophic. The linear model is a *reasonable* baseline — which is exactly why the interesting question is whether a **flexible** fit for tree size does better.

## 5.2 Logistic regression and the MLE (binarized-outcome aside)

The outcome is continuous, so a logistic GLM is not the native model. As a clearly-labelled **aside** to illustrate maximum-likelihood estimation and the log-odds interpretation, we binarize mortality into "high" (top tertile of log mortality) vs. the rest and fit a logistic GLM with the same predictors.

```r
d$hi <- as.integer(d$y > quantile(d$y, 2/3))      # 1 = high-mortality size-class
logit_fit <- glm(hi ~ lPN + llma + lwd + ldbh + nkT,
                 family = binomial(link = "logit"), data = d)
summary(logit_fit)$coefficients
```

```
##               Estimate Std. Error  z value     Pr(>|z|)
## (Intercept)  47.02867   22.48702  2.09137 0.0364949637
```

```
## lPN            0.91296    1.01116  0.90289 0.3665864689
## llma          -1.99496    0.57345 -3.47886 0.0005035551
## lwd          -14.14872    4.13051 -3.42542 0.0006138524
## ldbh          -1.58474    0.32667 -4.85118 0.0000012273
## nkT            1.50384    0.52155  2.88338 0.0039342850
```

```
## Log-likelihood = -61.89 | Null deviance = 178.68 (df 139) | Residual deviance = 123.79 (df 134)
↪  | AIC = 135.79
```

```
## log(DBH size) log-odds coefficient = -1.585  ->  odds ratio = 0.205 per unit of log-size
```

**MLE, likelihood, and log-odds.** Logistic regression has no closed-form least-squares solution; its coefficients are found by maximizing the Bernoulli **log-likelihood** $\ell(\beta) = \sum_i \left[y_i \log p_i + (1 - y_i)\log(1 - p_i)\right]$ with $p_i = 1/(1 + e^{-x_i^\top \beta})$, solved numerically by iteratively reweighted least squares. The fit reduces deviance from the null 178.7 to 123.8 (deviance $= -2\ell$; a drop means the predictors help). Each coefficient is a change in **log-odds**: the log-size coefficient of -1.585 means larger stems have 0.205× the odds of being a high-mortality class — smaller trees die faster, consistent with the negative OLS scaling exponent.

## 5.3 Is the size–mortality relationship linear? Polynomials

MST says log mortality should be **linear** in log size. We test that by adding quadratic and cubic terms in `ldbh` while holding the other predictors linear, and compare with nested $F$-tests.

```r
lm_lin  <- lm(y ~ lPN + llma + lwd + ldbh + nkT, data = d)
lm_pol2 <- lm(y ~ lPN + llma + lwd + poly(ldbh, 2, raw = TRUE) + nkT, data = d)
lm_pol3 <- lm(y ~ lPN + llma + lwd + poly(ldbh, 3, raw = TRUE) + nkT, data = d)
anova(lm_lin, lm_pol2, lm_pol3)
```

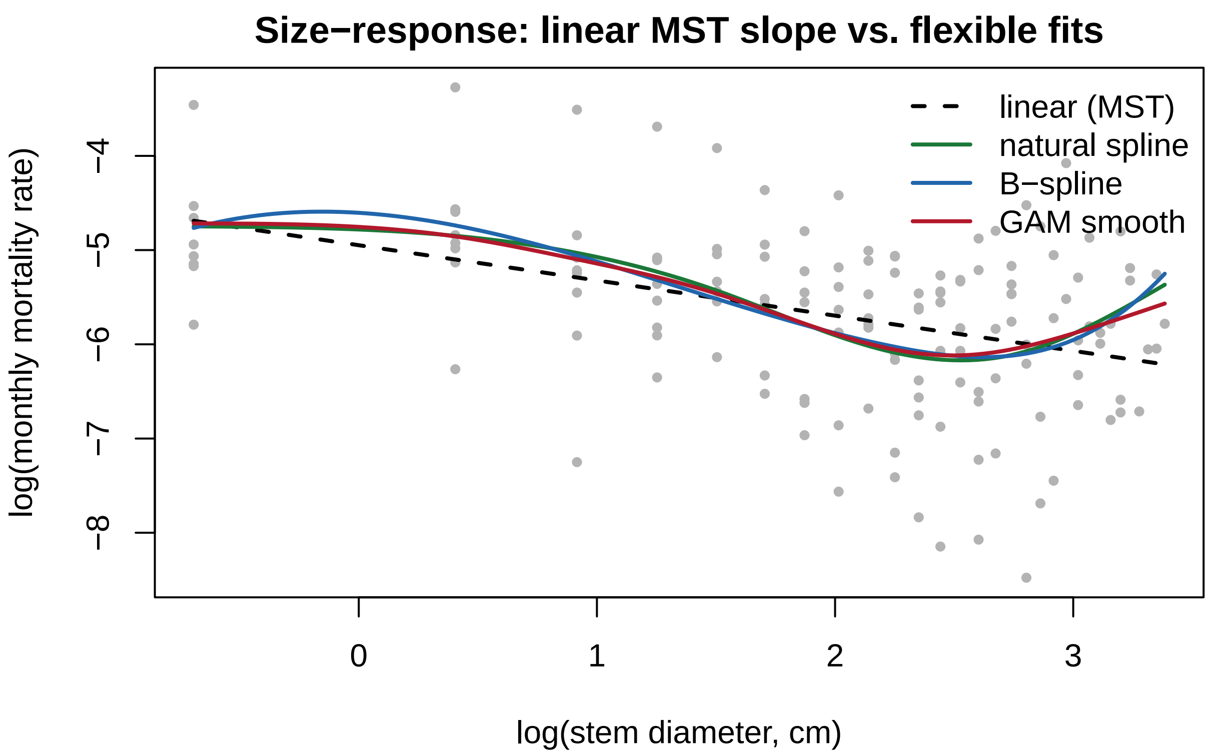
```
## Analysis of Variance Table
##
## Model 1: y ~ lPN + llma + lwd + ldbh + nkT
## Model 2: y ~ lPN + llma + lwd + poly(ldbh, 2, raw = TRUE) + nkT
## Model 3: y ~ lPN + llma + lwd + poly(ldbh, 3, raw = TRUE) + nkT
##   Res.Df  RSS Df Sum of Sq     F   Pr(>F)
## 1    134 69.0
## 2    133 67.8  1      1.15  2.52     0.11
## 3    132 60.2  1      7.64 16.76 0.000074 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Adding a single quadratic term is not significant, but the **cubic** term captures real structure (the size–mortality curve steepens and then flattens), which the linear MST slope averages over.

## 5.4 Splines: natural and B-splines on log size

Splines give the flexible fit a locally-adaptive shape without a global high-degree polynomial's tail wobble. We fit a natural cubic spline (`ns`) and a B-spline (`bs`) in `ldbh`, holding the other predictors linear, and plot the implied size–mortality curve.

```r
dbh_knots <- quantile(d$ldbh, c(.25, .5, .75))
dbh_bk    <- range(d$ldbh)
fit_ns <- lm(y ~ lPN + llma + lwd +
               ns(ldbh, knots = dbh_knots, Boundary.knots = dbh_bk) + nkT, data = d)
fit_bs <- lm(y ~ lPN + llma + lwd +
               bs(ldbh, knots = dbh_knots[2], Boundary.knots = dbh_bk) + nkT, data = d)
```




All three flexible fits agree: mortality falls steeply with size among small trees, then flattens for large trees — a curve the single straight line cannot follow.

## 5.5 GAM: a penalised smooth for size (and temperature)

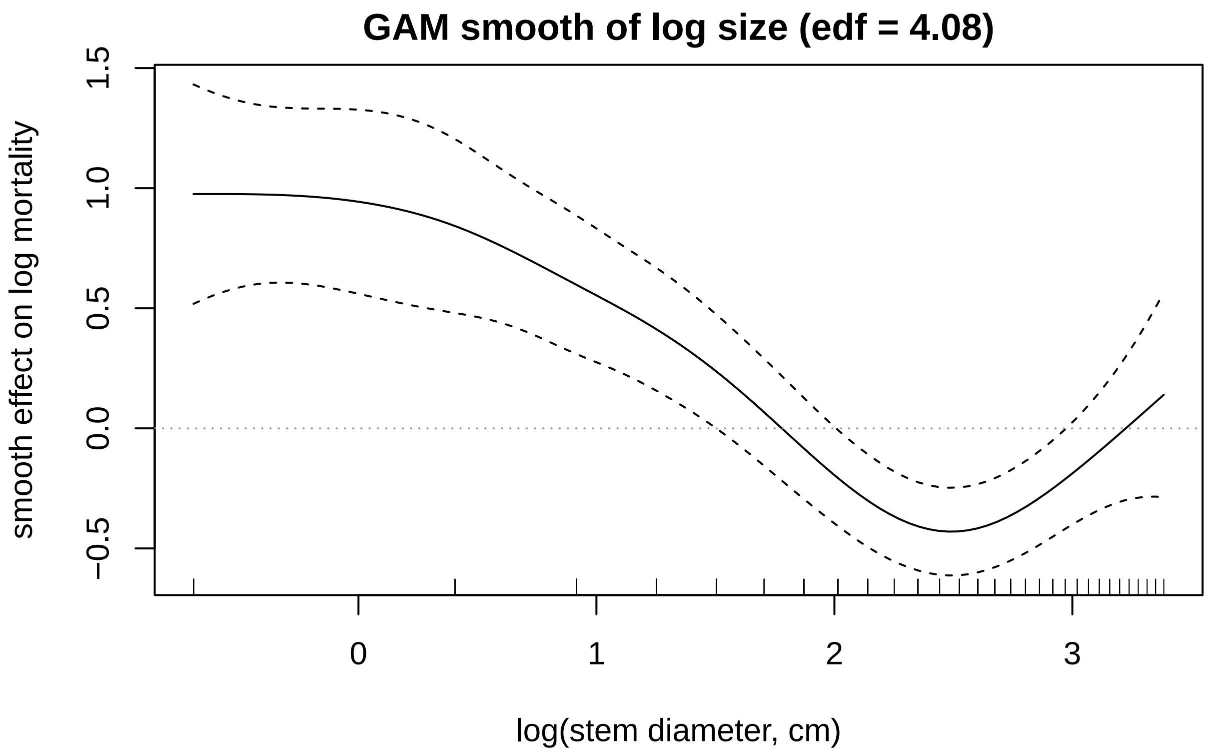
`mgcv::gam` chooses the smoothness of the size term automatically by penalised likelihood (GCV/REML), reporting the **effective degrees of freedom** (edf) — a data-driven measure of how nonlinear the term is (edf = 1 would mean "linear").

```
gam_dbh  <- gam(y ~ lPN + llma + lwd + s(ldbh) + nkT, data = d, method = "REML")
summary(gam_dbh)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## y ~ lPN + llma + lwd + s(ldbh) + nkT
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   26.694      5.181    5.15  9.3e-07 ***
## lPN            0.699      0.244    2.86   0.0049 **
## llma          -0.231      0.134   -1.72   0.0870 .
## lwd           -2.507      1.005   -2.49   0.0139 *
## nkT            0.825      0.109    7.55  6.7e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##          edf Ref.df    F p-value
## s(ldbh) 4.08   4.97 12.2  <2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.485   Deviance explained = 51.4%
## -REML = 150.28  Scale est. = 0.45152   n = 140
```

```r
par(mar = c(4.2, 4.2, 2, 1))
plot(gam_dbh, select = 1, shade = TRUE, seWithMean = TRUE,
     xlab = "log(stem diameter, cm)", ylab = "smooth effect on log mortality",
     main = sprintf("GAM smooth of log size (edf = %.2f)", summary(gam_dbh)$edf[1]))
abline(h = 0, col = "grey60", lty = 3)
```




The size smooth uses edf ≈ 4.08 with $p < 0.001$ — strong, statistically supported curvature in the size–mortality relationship, beyond the single MST exponent. (Growing-season temperature enters through only eight discrete site values, so it is kept as the linear Arrhenius term $-1/kT$ rather than a poorly-identified smooth; a version with a small-basis temperature smooth `s(GST, k=5)` gives essentially the same held-out error, below.)

## 5.6 Out-of-sample comparison (the deciding test)

We compare every model with **common 10-fold cross-validation** on the same folds, reporting held- out RMSE (lower is better). Splines use fixed full-data knots so the basis is identical across folds.

```r
K <- 10; n <- nrow(d)
folds <- sample(rep(1:K, length.out = n))

cv_rmse <- function(fitfun, predfun) {
  se <- numeric(0)
  for (k in 1:K) {
    tr <- d[folds != k, ]; te <- d[folds == k, ]
    m  <- fitfun(tr)
    se <- c(se, (te$y - predfun(m, te))^2)
  }
  sqrt(mean(se))
```

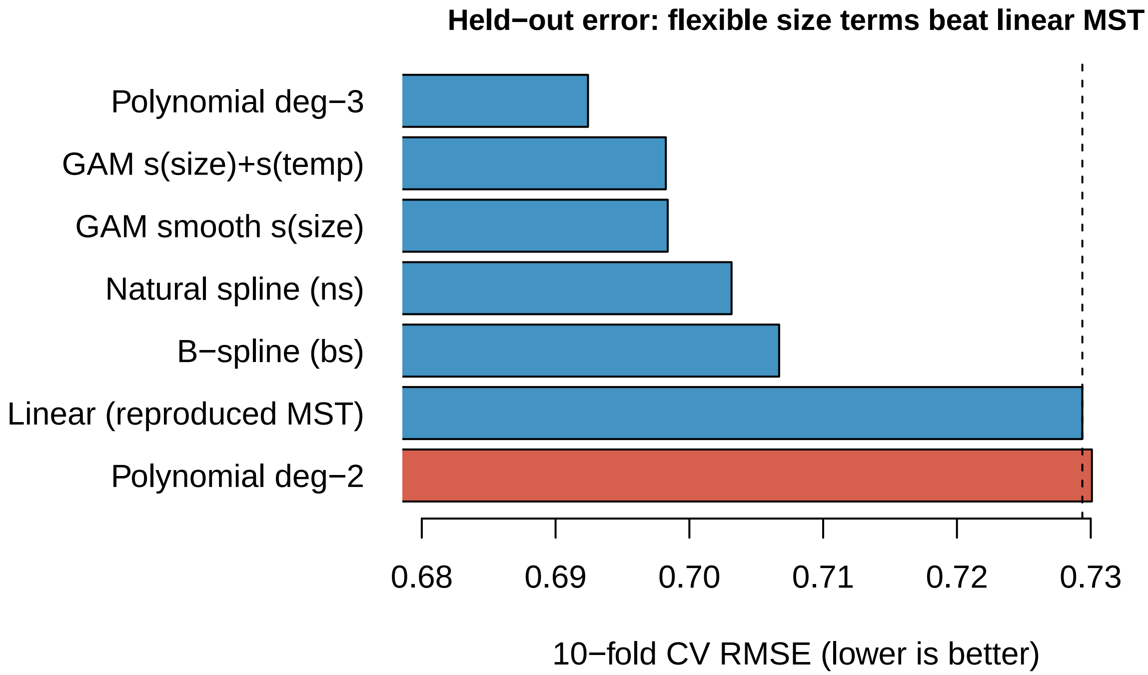
```r
}

# helper predictors for readability
p_lm  <- function(m, te) predict(m, te)
p_gam <- function(m, te) as.numeric(predict(m, te))
ns_f  <- function(x) ns(x, knots = dbh_knots, Boundary.knots = dbh_bk)
bs_f  <- function(x) bs(x, knots = dbh_knots[2], Boundary.knots = dbh_bk)

res <- c(
  "Linear (reproduced MST)" =
    cv_rmse(function(tr) lm(y ~ lPN+llma+lwd + ldbh + nkT, tr), p_lm),
  "Polynomial deg-2" =
    cv_rmse(function(tr) lm(y ~ lPN+llma+lwd + poly(ldbh,2,raw=TRUE) + nkT, tr), p_lm),
  "Polynomial deg-3" =
    cv_rmse(function(tr) lm(y ~ lPN+llma+lwd + poly(ldbh,3,raw=TRUE) + nkT, tr), p_lm),
  "Natural spline (ns)" =
    cv_rmse(function(tr) lm(y ~ lPN+llma+lwd + ns_f(ldbh) + nkT, tr), p_lm),
  "B-spline (bs)" =
    cv_rmse(function(tr) lm(y ~ lPN+llma+lwd + bs_f(ldbh) + nkT, tr), p_lm),
  "GAM smooth s(size)" =
    cv_rmse(function(tr) gam(y ~ lPN+llma+lwd + s(ldbh) + nkT, data=tr), p_gam),
  "GAM s(size)+s(temp)" =
    cv_rmse(function(tr) gam(y ~ lPN+llma+lwd + s(ldbh) + s(GST,k=5), data=tr), p_gam)
)
```

Table 9: 10-fold cross-validated held-out RMSE for the reproduced linear model and each Day-2 extension (same folds). Negative percentages beat the linear model.

| Model | CV RMSE | vs. linear |
|---|---|---|
| Linear (reproduced MST) | 0.7294 | +0.0% |
| Polynomial deg-2 | 0.7301 | +0.1% |
| Polynomial deg-3 | 0.6924 | -5.1% |
| Natural spline (ns) | 0.7032 | -3.6% |
| B-spline (bs) | 0.7067 | -3.1% |
| GAM smooth s(size) | 0.6984 | -4.2% |
| GAM s(size)+s(temp) | 0.6982 | -4.3% |

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## 6 Takeaway

Reproducing the paper's headline OLS is exact ($R^2 = 0.43$, all five MST/trait terms significant), and it is a genuinely good linear model. But Day-2 flexibility earns its keep on the term the *theory* is most specific about: a GAM smooth of log stem size uses edf $\approx 4.08$ ($p < 0.001$), and the cubic-polynomial, spline, and GAM fits all cut held-out RMSE by roughly 5% relative to the straight MST line — because mortality drops sharply among small trees and then plateaus, curvature a single scaling exponent cannot express. The lesson is the Day-2 lesson: when a theory predicts a functional form, splines and GAMs are the honest way to check it, and here they reveal that the size–mortality scaling is **not** a single power law but flattens in large trees — an out-of-sample-validated departure from the MST straight line.

```
## R 4 6.0 | mgcv 1.9.4 | splines (base)
```